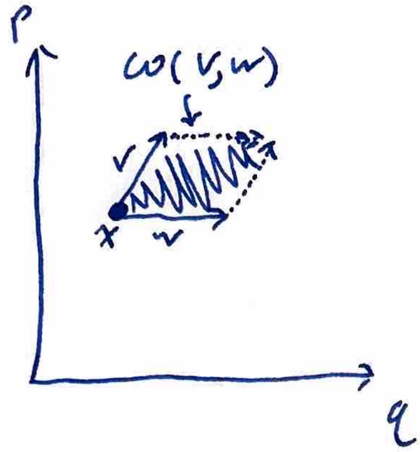
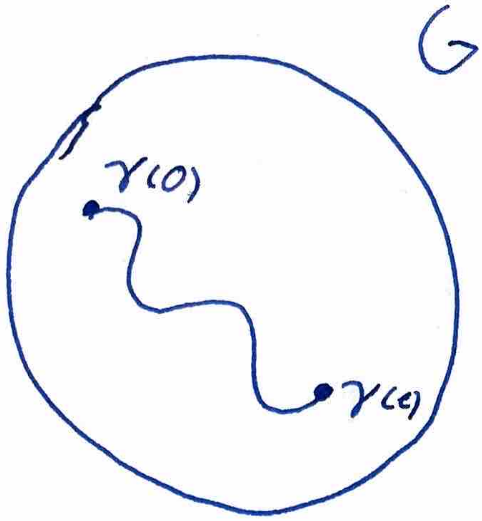
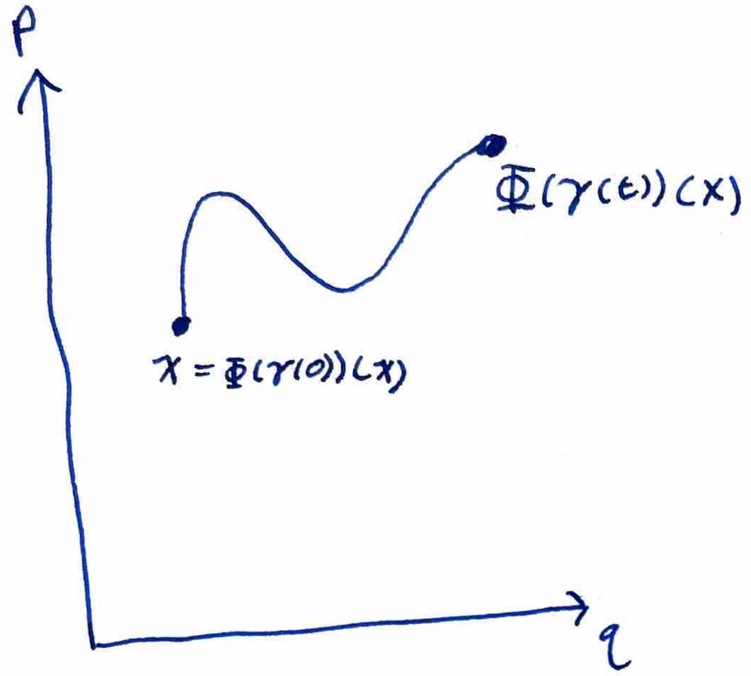
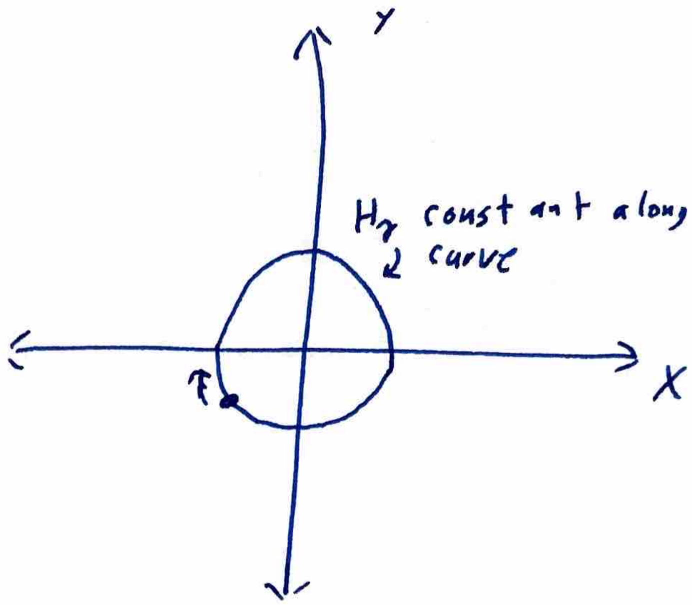
- Geometric mechanics: using tools from geometry to simplify problems in classical mechanics
- Phase space P: represents space of possible states of a physical system; all possible positions and momenta of the system
- Reducing phase space makes computations more simple. Symplectic reduction does that algebraically.
- P is often smooth, and has additional structure: the symplectic form ω.
  - ω takes two vectors tangent to P at the same point and gives a real number
  - Linear in both arguments
  - Alternating $\omega(v, w) = -\omega(w, v)$
  - Non-degenerate $\omega(v, w) = 0 \forall w \implies v = 0$
  - Does not change when you move in space
  - Example: (signed) area of a parallelogram
  - Encodes rules of motion of classical mechanics
- A symmetry of the system should not change the symplectic form.
- A symmetry can be seen as a group action.
  - A group G acting on P gives a homomorphism $\Phi : G \to \mathrm{Aut}(P)$
  - Symmetries defined the group action are the automorphism that leave the symplectic form unchanged.
  - Such automorphisms are called symplectomorphisms; they form a group $\mathrm{Symp}(P, \omega)$
  - Example: $O(n)$ is group of distance-preserving transformations of Euclidean n-space that also preserve the origin; this is same thing, but instead preserving symplectic form rather than distance metric
- We are only interested in groups whose action-induced homomorphism sends each element to a symplectomorphism. Moreover, these groups should also have the structure of a smooth space (Lie group)
- Let $\gamma : \mathbb{R} \to G$ be a continuous group homomorphism. The image of a group homomorphism gives a subgroup of the G, so γ defines what we call a one-parameter subgroup, which defines a continuous curve in G.
- The action gives a continuous family of symplectomorphisms $\Phi(\gamma(t)) : P \to P$. If you take a point $x \in P$, and apply the symplectomorphisms $\Phi(\gamma(t))$ by varying t, then a curve is drawn in P.
- Noether's theorem: there exists a function $H_\gamma : P \to \mathbb{R}$ that is constant along orbits of $\Phi(\gamma(t))$, which are those curves drawn in P; H_γ is called a conserved quantity associated to γ.
  - Example: angular momentum constant in a circular orbit obtained by repeatedly rotating a phase-space point via the symplectomorphism coming from the rotation group.

- The momentum map J takes a point in P and maps it to a function that assigns a conserved quantity to each one-parameter subgroup: $J(p)(\gamma) = H_\gamma(p)$. Convenient way to store all the conserved quantities in a single object.
- If we choose one of these functions μ, look at all states having exactly those conserved quantities; this is a subset $J^{-1}(\mu) \subset P$.
  - Sard's theorem from differential topology tells us that (almost) all values of μ give $J^{-1}(\mu)$ to be a smooth space.
  - Since G acts on P and preserves the momentum map J, it acts on $J^{-1}(\mu)$.
  - We can form the reduced phase space $P_{red} = J^{-1}(\mu)/G$, which is also smooth.
  - This smooth space has a symplectic form of its own that's obtained naturally from (P, ω).
  - $\dim P_{red} = \dim J^{-1}(\mu) - \dim G$
- Example: particle in 2D attached to a spring at the origin
  - 4D phase space (2 position coordinates, 2 momentum)
  - Rotation symmetry (rotating both position and momentum vectors by some angle is a symplectomorphism)
  - Under rotation, momentum map gives angular momentum
  - Fix some nonzero angular momentum L. The constraint $xp_y - yp_x = L$ is one smooth equation in 4 variables, so it cuts the 4D phase space into a 3-dimensional hypersurface $J^{-1}(L)$
  - Each rotation orbit is 1D, so when we quotient out the rotation group we're collapsing every circle orbit to a single point.
  - Now we have a 2D phase space!
- Summary: symmetry simplifies mechanics.


p
ω(v,w)
v
x
w
q


G
γ(0)
γ(t)


p
Φ(γ(t))(x)
x = Φ(γ(0))(x)
q


Y
Hγ constant along
curve
X